CNN-BASED CLASSIFICATION OF HYPERSPECTRAL POLYMER IMAGES

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Submitted by	Raghav Bansal
Roll Number	2401ME51
Department	Mechanical Engineering

Abstract

This project develops a 2D convolutional neural network (CNN) to classify hyperspectral images of four polymers: HDPE, PET, PP and PS. The 168-band BIL/HDR images were inspected, partitioned by acquisition group, reduced to a common set of 10 PCA-selected original bands, normalized using training data, and padded to a fixed spatial size. Three CNN configurations were trained and evaluated. Each achieved 100% accuracy on the 30-image held-out test set; the baseline model had the lowest validation loss and was selected as the final model. The small test-set size is an important limitation when interpreting this result.

Project workflow

BIL/HDR data → 168-band hyperspectral cube → common 10-band PCA selection → training-only normalization → padding to 68 × 73 × 10 → CNN training and evaluation → polymer prediction.

1. Introduction and Dataset

Hyperspectral imaging records a spectrum at each pixel. Unlike an RGB image, it contains many narrow wavelength bands that can help distinguish materials through their spectral response. The supplied dataset was acquired using a Resonon Pika IR line-scan system and contains four polymer classes.

Item	Dataset information
Classes	HDPE, PET, PP and PS
Total images	200 (50 per class)
Spectral bands	168 bands; nominal range 900–1700 nm
Image dimensions	Height: 26–68 pixels; width: 27–73 pixels
File format	BIL image files with corresponding HDR metadata

2. Data Partition and Preprocessing

The 200 images were divided into training (140; 70%), validation (30; 15%) and test (30; 15%) subsets using a fixed random seed. Images belonging to the same acquisition scan group were kept in the same subset to reduce data leakage. Because groups contain multiple related fragments, the validation and test class counts are not perfectly balanced: the test set contains 10 HDPE, 10 PET, 5 PP and 5 PS images.

PCA was fitted using representative spectra from training images only. Three principal components retained 95.01% cumulative explained variance. Original bands were ranked using variance-weighted squared loadings, and one common set of 10 original bands was selected: 159–168. The same indices were applied to every split and during inference.

Per-band means and standard deviations were calculated from training pixels only. The selected-band images were normalized and symmetrically padded to 68 × 73 × 10. No cropping, resizing or interpolation was applied. The saved preprocessing configuration is reused during inference; PCA and normalization are not refitted for a new image.

3. CNN Design and Experiments

A 2D CNN was designed for an input size of 68 × 73 × 10 and four output classes. The baseline network uses three convolution blocks with 16, 32 and 64 filters, respectively, with batch normalization and ReLU activations. Max-pooling provides downsampling; global average pooling and a four-unit fully connected classification head produce class scores. Training used Adam, cross-entropy loss, mini-batches of 16 and a maximum of 40 epochs, with validation monitoring.

Configuration	Controlled change	Best validation loss
Baseline CNN	Three convolution blocks; no dropout	0.013672
Dropout CNN	Baseline-style network + dropout (0.30)	≈ 0.01972
Shallow CNN	Reduced to two convolution blocks	0.066161

The baseline had the lowest validation loss. The dropout experiment did not improve validation performance, while the shallower network had a substantially higher validation loss. These results indicate that, for this dataset and training setup, the baseline learned the training task most effectively.

4. Test Results

All three models correctly classified all 30 test images. The resulting confusion matrix for each model is identical and has no off-diagonal errors.

True \ Predicted	HDPE	PET	PP	PS
HDPE	10	0	0	0
PET	0	10	0	0
PP	0	0	5	0
PS	0	0	0	5

Model	Accuracy	Mean recall	Macro precision	Macro F1	Weighted F1
Baseline	1.00	1.00	1.00	1.00	1.00
Dropout CNN	1.00	1.00	1.00	1.00	1.00
Shallow CNN	1.00	1.00	1.00	1.00	1.00

For every class and every model, precision, recall and F1-score were 1.00. The highest precision and highest recall are therefore tied across all four classes. No class pair was confused in the test set, so there are no observed spectral or visual confusion cases to analyze from this test sample.

5. Required Evaluation Questions

1. Which class has the highest precision and recall? All four classes are tied: each has precision = 1.00 and recall = 1.00 on the test set.

2. Which classes are most frequently confused? None were confused in the 30 test predictions. Similar spectral responses, overlapping absorption features, sample surface variation or acquisition differences could cause confusion in a larger or more varied dataset, but no such error was observed here.

3. Can accuracy be high while one class has poor recall? Yes. If a model correctly classifies most samples from common classes but misses many samples from a less represented class, overall accuracy can remain high while that class's recall is low. The confusion matrix exposes these false negatives.

4. When is macro F1 more informative than accuracy? Macro F1 gives equal weight to each class, so it is more informative when class counts are imbalanced or performance on every polymer matters equally. Accuracy can be dominated by the more frequent classes.

5. Which metric matters most for automated sorting? Precision is important when false positives contaminate a recovered polymer stream. Recall is important when false negatives cause recoverable material to be rejected. The operational priority depends on the relative cost of contamination and material loss; F1 is useful when a balance is required. A practical sorting line should monitor both precision and recall rather than relying on accuracy alone.

6. Unseen-Image Demonstration and Limitations

A MATLAB prediction routine was implemented to accept a BIL/HDR image pair, apply the saved common band indices, training-derived normalization parameters and fixed padding, and predict one of the four polymer classes using the trained baseline network. PCA is not refitted and the CNN is not retrained during inference.

The demonstration performed so far used an image from the held-out test set. It demonstrates that the inference pipeline runs, but it is not an independent external unseen-specimen evaluation. A separate image from outside the 200-image dataset would provide stronger evidence for the assignment's unseen-image objective.

The test set contains only 30 images, and the perfect test score should be interpreted cautiously. Group-wise partitioning was used to keep related scan fragments together, but the limited sample size and possible acquisition-specific similarities restrict claims about generalization to new specimens, scanners or acquisition conditions. Further validation on independent specimens is recommended.

7. Conclusion

A complete hyperspectral polymer-classification pipeline was implemented, including BIL/HDR reading, group-wise data partitioning, training-only PCA band selection, normalization, padding, CNN training, evaluation and image-level inference. Three CNN configurations were compared. All achieved 100% accuracy on the 30-image test set, while the baseline CNN achieved the lowest validation loss (0.013672) and was selected as the final model. The results are promising for this dataset, but independent-specimen testing is required before claiming reliable deployment in an automated sorting system.

Key implementation outputs

Output	Purpose
Model/trained_cnn.mat	Selected baseline network
Results/preprocessing_config.mat	Saved preprocessing settings
Results/pca_band_selection.mat	PCA and selected-band information
Results/all_models_test_summary.csv	Overall model comparison
Results/all_models_test_class_metrics.csv	Class-wise test metrics
Results/Plots/	PCA, training and evaluation plots

Reference: ME3101-Autumn2026, Mini-Project 2 assignment brief, “CNN-Based Classification of Hyperspectral Polymer Images.”